MATH329 · OPERATIONS RESEARCH

Midterm 471 Solutions

60 minutes

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01 Solution Q1: True / False

#	Answer	Reason
1	**F**	The model's parameters are known. The unknowns are the variables.
2	**T**	This is the definition of a BFS.
3	**T**	This is the definition of degeneracy.
4	**F**	If all the ratios are negative or infinite, no variable can leave and the solution is unbounded.
5	**F**	The solution must satisfy the constraints.

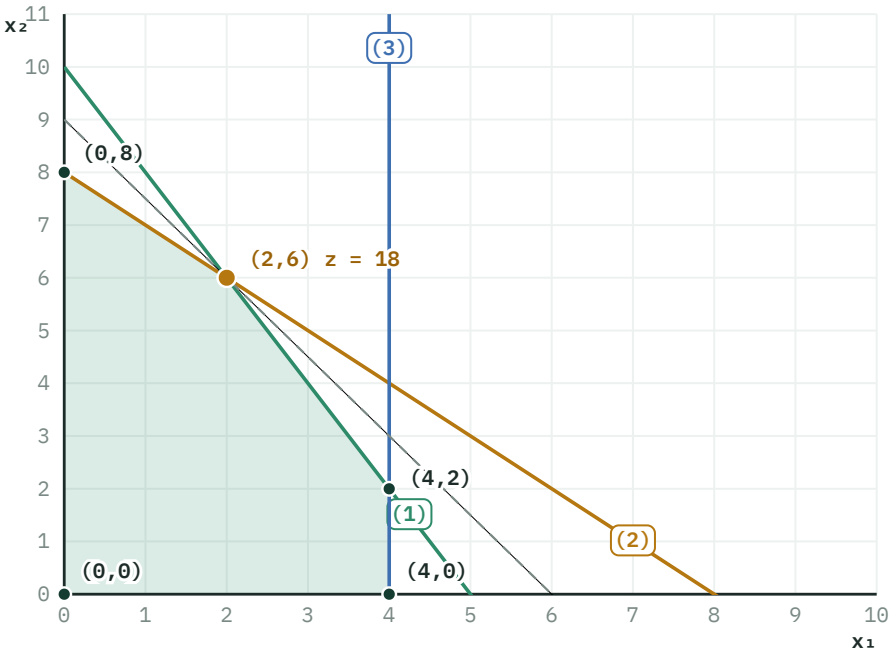
02 Solution Q2: graphical

1 Prepare the constraints

(1) $2x_1 + x_2 = 10$ points (0, 10) and (5, 0)

(2) $x_1 + x_2 = 8$ points (0, 8) and (8, 0)

(3) $-x_1 \geq -4 \times (-1) \rightarrow x_1 \leq 4$ vertical line at $x_1 = 4$



2 The corner points

Corner	Where it comes from	$z = 3x_1 + 2x_2$
(0, 0)	origin	0
(4, 0)	(3) with the x_1 axis	12
(4, 2)	(1) with (3)	16
(2, 6)	(1) with (2)	**18**
(0, 8)	(2) with the x_2 axis	16

(1) − (2): $x_1 = 2$ $\quad 2 + x_2 = 8 \quad \rightarrow \quad x_2 = 6$

Objective Value = 18 at $x_1 = 2$ and $x_2 = 6$

03 Solution Q3: standard form and simplex

a Standard form

$$\text{Maximize } Z = 2x_1 + 3x_2 + 0s_1 + 0s_2$$

$$\text{S.to} \quad x_1 + 2x_2 + s_1 = 16$$

$$2x_1 + x_2 + s_2 = 18$$

$$x_1, x_2, s_1, s_2 \geq 0$$

b Simplex

Basic	x_1	x_2	s_1	s_2	Sol.	Ratio
Z	−2	−3	0	0	0	
s_1	1	**2**	1	0	16	8
s_2	2	1	0	1	18	18
Z	−1/2	0	3/2	0	24	
x_2	1/2	1	1/2	0	8	16
s_2	**3/2**	0	−1/2	1	10	20/3
Z	0	0	4/3	1/3	82/3	
x_2	0	1	2/3	−1/3	14/3	
x_1	1	0	−1/3	2/3	20/3	

Iteration 1: `x2` enters because `−3` is the most negative. The ratios are `16/2 = 8` and `18/1 = 18`, so `s1` leaves and the pivot is `2`.

Iteration 2: `−1/2` remains under `x1`, so it enters. The ratios are `8/(1/2) = 16` and `10/(3/2) = 6.67`, so `s2` leaves and the pivot is `3/2`.

Optimum z = 82/3 = 27.33 at x_1 = 20/3 = 6.67 and x_2 = 14/3 = 4.67

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Check:  z = 2(20/3) + 3(14/3) = 40/3 + 42/3 = 82/3
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The key writes decimals: `1.33`, `0.33`, `27.33`. Its marking: three marks for each table and three for the optimum, with half a mark off for each arithmetic slip.